

Policy interactions reshape the outcomes of carbon pricing policies

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The adoption and effectiveness of carbon pricing are highly reshaped by interactions with other climate mitigation policies. A global comparative assessment of policy synergies and conflicts can guide policymakers in designing policy portfolios that can achieve higher mitigation cost-effectiveness.

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The policy problem

Countries are struggling to meet the Paris Agreement climate targets despite an increasing number of climate policies. Carbon pricing – through taxes or emissions trading – is theoretically the most cost-effective tool for reducing emissions; yet, its real-world performance varies substantially across countries. A central reason is that carbon pricing rarely operates in isolation. It is often implemented in conjunction with other climate policies, producing synergies or conflicts depending on the policy design, institutional settings and implementation contexts. With 51 countries now implementing carbon pricing, approximately 28% of global emissions are affected. In the absence of evidence-based guidance on policy mix design, there is a substantial risk that the mitigation potential of carbon pricing will not be fully realized. Without a systematic understanding of these interactions, governments risk deploying ‘cluttered’ portfolios that weaken price signals, drain public coffers and undermine market credibility.

The findings

We find that carbon pricing reduces emissions intensity by 15.4% for emissions trading systems and 8.5% for carbon taxes, but policy interactions systematically affect these outcomes (Fig. 1). For emissions trading systems in particular, effectiveness depends on whether accompanying policies support market functioning and price signals or instead create competing incentives that weaken them. Aligning subsidies and information policies with different maturity levels of carbon markets, while using public investment to ease adjustment costs, leads to more coherent and effective mitigation outcomes. Simulation results show that reducing policy conflicts can substantially amplify the impact of carbon pricing, raising average emission intensity reductions from about 10% to nearly 22%. Before implementing carbon pricing policies, policymakers should assess potential conflicts or synergies with existing policy foundations. Our findings highlight the need for dynamic policy evaluation rather than static assessments – what works changes as markets mature and circumstances evolve.

The study

This study drew on a global-scale dataset that integrates multiple detailed sources on carbon pricing and other climate policies, covering more than 10,000 policies across more than 100 countries and providing unusually broad and comprehensive coverage. To measure carbon pricing effectiveness, we used a synthetic control method to construct counterfactual outcomes for target countries, allowing us to estimate what would have happened without carbon pricing policies. We then developed a Global Climate Policy Index that quantifies the strength, coverage and legal enforceability of different policy types, making otherwise heterogeneous national policy environments structured and comparable. We then estimated how these policies interact with carbon pricing and affect the estimated policy effects. Our analysis obtained highly policy-relevant results about climate policy interactions, if not a strict causal result, with massive and extensive robustness checks, providing global evidence to guide climate policy design under different policy foundations.

Recommendations for policy

- Policymakers should evaluate both synergistic and conflicting interactions to ensure the robust performance of carbon pricing instruments
- Carbon pricing designs – including coverage, mechanism and intensity – should be aligned with the pre-existing policy mix to maximize mitigation effectiveness
- In emerging markets, government research and development and infrastructure provision can anchor price signals and boost the development of carbon markets
- Mature market portfolios should be streamlined to prevent complementary policies from distorting price formation or undermining market credibility

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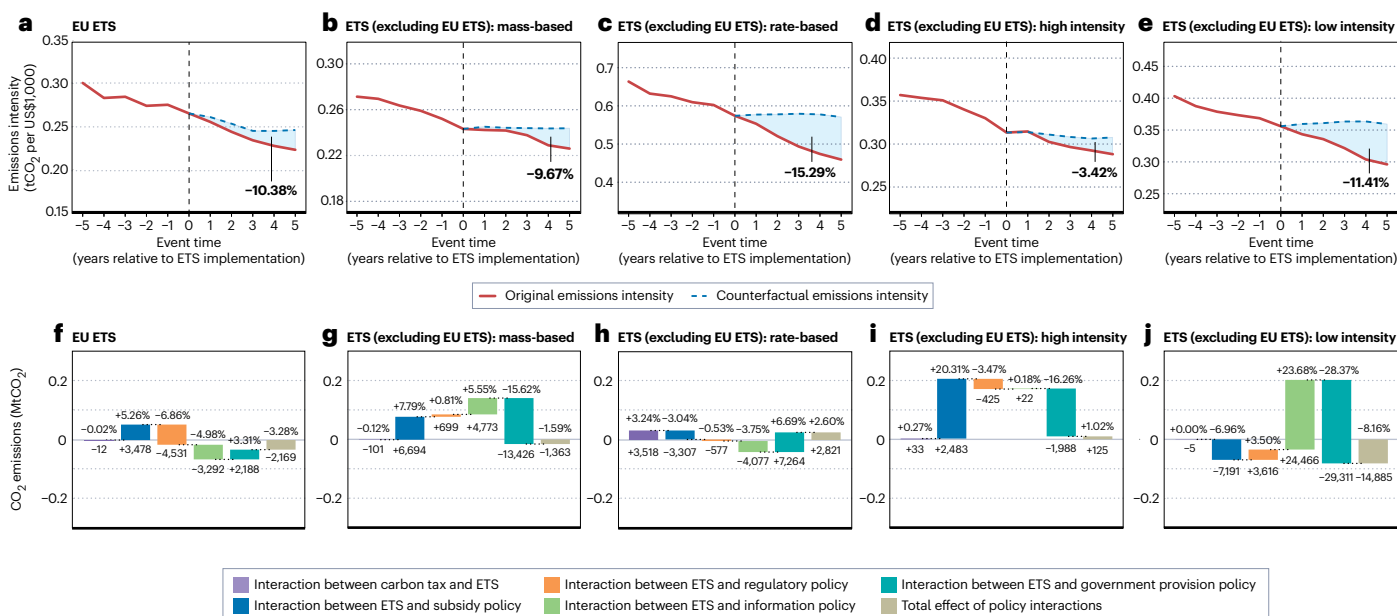


Fig. 1 | Direct and decomposed policy effects and interactions with emissions trading systems. **a–e**, The event study of treatment effects from emissions trading systems (ETSs) includes time-fixed effects. **a**, The event study of treatment effects for the EU ETS, including time-fixed effects. **b**, The event study results for mass-based ETSs. **c**, Results for rate-based ETSs. **d**, High-intensity ETSs. **e**, Low-intensity ETSs. In **b–e**, the EU ETS is excluded. Original and counterfactual emissions intensity trajectories depict realized and hypothetical emissions paths following policy implementation, respectively. Because there are overlaps between the carbon tax and trading, a 5-year event study result is presented to showcase the policy effect. Hence, the reduction percentage is slightly different from the full sample. The year when a country first implemented carbon pricing policies is set as the first period and the previous year as the zeroth period. The

values in the graphs represent the change in the weighted average of emissions intensity for all periods with and without policies, thus reflecting the aggregated effect. **f–j**, Changes in CO₂ emissions caused by each interaction effect, except for the time effect. **f**, The decomposed interaction effects for the EU ETS. **g**, The interaction effects for mass-based ETSs. **h**, The interaction effects for rate-based ETSs. **i**, The interaction effects for high-intensity ETSs. **j**, The interaction effects for low-intensity ETSs. In **g–j**, the EU ETS is excluded. Each bar in the graphs indicates the magnitude of the change in CO₂ emissions caused by the interaction of the corresponding policy during 1996–2019, with corresponding percentage (above) and absolute emission values (below; measured in million tonnes). Figure adapted from L. Wu et al. *Nat. Clim. Change* <https://doi.org/10.1038/s41558-026-02574-4> (2026), Springer Nature Limited.

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Further reading

- Stechemesser, A. et al. Climate policies that achieved major emission reductions: global evidence from two decades. *Science* **385**, 884–892 (2024).
This paper identifies which climate policies and policy mixes have delivered large, sustained emission reductions across countries, highlighting the importance of effective policy design.
- Wu, L., Huang, Z., Zhang, X. & Wang, Y. Harmonizing existing climate change mitigation policy datasets with a hybrid machine learning approach. *Sci. Data* **11**, 580 (2024).
This paper introduces a harmonized global climate policy dataset, providing a standardized foundation for large-scale comparative policy analysis.
- Acemoglu, D., Aghion, P., Bursztyn, L. & Hémous, D. The environment and directed technical change. *Am. Econ. Rev.* **102**, 131–166 (2012).
This paper provides a foundational theoretical framework showing how climate policies, especially carbon pricing, can redirect innovation towards clean technologies and shape long-run mitigation pathways.

- van den Bergh, J. et al. Designing an effective climate-policy mix: accounting for instrument synergy. *Clim. Policy* **21**, 745–764 (2021).
This paper synthesizes theoretical and empirical evidence on climate policy interactions, providing clear principles for designing effective and politically feasible policy mixes.
- Meckling, J. & Allan, B. B. The evolution of ideas in global climate policy. *Nat. Clim. Change* **10**, 434–438 (2020).
This paper traces the shift in global climate policy from a focus on carbon pricing to broader policy mixes, explaining why policy interactions and coherence have become central to effective mitigation.

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Competing interests

The authors declare no competing interests.